Swopnab Bikram Karki

+1 986-600-8163 | swopnabbikram@gmail.com | linkedin.com/in/swopnabbkarki | github.com/Swopnab

Education

University of Texas at Arlington — Arlington, TX
B.S. in Computer Science — *Expected Graduation: Fall 2028*

TechnicalSkills

Languages: Python, Java, C++, JavaScript, TypeScript, SQL

Web & Backend: React, FastAPI, Flask, Hono, Node.js, REST APIs, HTML5, CSS3, Canvas

AI & Vision: Ollama, RAG, embeddings, MediaPipe

Data & Cloud: SQLite, SQLAlchemy, FTS5/BM25, indexing, transactions, Cloudflare Workers/D1, AWS (S3, IAM)

Security: JWT, RBAC, refresh-token rotation, PBKDF2-SHA256, session revocation, user-data isolation

Browser Storage: IndexedDB, Service Workers, offline caching

Testing & Tools: pytest, Playwright, unit/integration testing, performance benchmarking, Git, GitHub Actions, CI/CD

Projects

WorkLikeADog - AI Job Search Companion 2026
Local AI Workflow Platform — *Python, FastAPI, SQLAlchemy, Ollama, Playwright, LaTeX*

- Validated eligibility and restart recovery with 73 passing tests by adding profile-based graduation checks, skill-boundary rules, and persistent recovery regressions.
- Kept duplicate-check p95 below 0.7 ms at 10,000 synthetic records by indexing normalized job identities.

SwopMobile - JWT Authentication & RBAC 2025-2026
Serverless Authentication Service — *JavaScript, Cloudflare Workers, Hono, D1*

- Hardened authentication with 16 passing tests by implementing atomic token rotation and live-session revocation.
- Enforced 3 permission roles by combining JWT session checks, RBAC, and SQL ownership filters.

CyberSteer - Hand-Controlled Racing Game 2026
Computer Vision / Browser Game — *JavaScript, HTML5 Canvas, MediaPipe Hands, CSS3*

- Enabled 3-lane hand steering by mapping MediaPipe palm coordinates to mirrored zones with hysteresis.
- Verified mobile layouts across 9 emulated viewports by testing touch, keyboard, rotation, and camera failures.

Music Player - Offline Audio Library 2026
Browser Audio / Local-First Library — *React, TypeScript, IndexedDB, HTML5 Audio*

- Reduced DOM elements by 95.5% at 1,000 synthetic tracks by paginating lists and disc carousels.
- Enabled imported-audio playback after an offline reload by caching the app shell and storing files in IndexedDB.

Zabum AI - Local AI Assistant 2026
Persistent Memory / Document Retrieval — *Python, Flask, SQLite FTS5, Ollama, PyTorch*

- Reduced retrieval p95 from 36.8 ms to 0.3 ms on 10,000 synthetic chunks by replacing full scans with an FTS5/BM25 shortlist and bounded reranking.
- Verified persistence and ingestion with 9 tests by using SQLite transactions and atomic document indexing.

Certifications

Microsoft Azure Essentials Professional Certificate by Microsoft and LinkedIn

Career Essentials in Software Development by Microsoft and LinkedIn

Excel Essential Training (Microsoft 365)